

- (c) Met $h = R/6$, toon aan dat $v_0 \sin \theta_0 = \sqrt{gR/3}$.

With $h = R/6$, show that $v_0 \sin \theta_0 = \sqrt{gR/3}$.

(2)

- (d) Met die reikwydte $R = \frac{v_0^2 \sin 2\theta_0}{g}$ en die gegewens in (c), toon aan dat

$$v_0 \cos \theta_0 = \frac{1}{2} \sqrt{3gR} \text{ indien } \sin 2\theta = 2 \sin \theta \cos \theta.$$

With the range $R = \frac{v_0^2 \sin 2\theta_0}{g}$ and the formula given in (c), show that

$$v_0 \cos \theta_0 = \frac{1}{2} \sqrt{3gR} \text{ if } \sin 2\theta = 2 \sin \theta \cos \theta.$$

- (e) Toon aan dat die tyd wat nodig is om die piekhoogte te bereik gegee word deur $t_p = \sqrt{R/3g}$.
Show that the time it takes to reach the top is given by $t_p = \sqrt{R/3g}$.

(3)